

HYUN JAE STEPHEN CHU

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RESEARCH INTERESTS

Bayesian Econometrics, Macroeconomics, International Finance, Machine Learning, Monetary and Fiscal Policy

EDUCATION

2025– Ph.D., Economics, *University of Notre Dame*
2024 M.A., Economics, *Korea University*
 Thesis Advisor: Kyu Ho Kang
2022 B.A., Economics and Financial Engineering, *Korea University*

WORKING PAPERS

“Time-varying Bayesian Variable Selection for Macroeconomic Forecasting: A Bayesian Regression Tree Approach”, with Kyu Ho Kang and Jaeho Kim [[SSRN](#)] [[Codes](#)]

Abstract: We propose a Bayesian Dirac Classification and Regression Tree (B-DART) model for macroeconomic forecasting that jointly estimates a tree-based regime configuration and selects predictors separately within each regime. By allowing local predictor sets to differ across terminal nodes, the model accommodates changes in predictor relevance across economic conditions. In an application to eight U.S. macroeconomic variables, B-DART delivers the most accurate forecasts in more target-horizon combinations than the Bayesian CART without local variable selection model. Its relative forecasting performance is particularly strong during the volatile post-COVID-19 period. Variable selection also affects the estimated regime structure where B-DART yields approximately five regimes on average, whereas Bayesian CART estimates a single regime for most target variables. These findings suggest that regime determination and within regime variable selection are mutually reinforcing.

PROFESSIONAL EXPERIENCE

2025 **National Assembly Budget Office (NABO)**
 Research Analyst, Population and Strategy Analysis Division, February 2025 - June 2025

RESEARCH EXPERIENCE

Korea University

2023–24 *Research Assistant for Kyu Ho Kang*

Projects:

Outlook of the Semi-conductor Industry using LBVAR (with SK Hynix)

Bank of Korea Jeju Island Tourism Business Index Development

NH Bank Warning Index (NHWI) Development

Bank of Korea Econometrics Data Analysis Library Development

TEACHING EXPERIENCE

University of Notre Dame

2026 (Fall) *Teaching Assistant: Macroeconomic Theory I (Graduate), Zachary Stangebye*

Korea University

2024 (Spring) *Teaching Assistant: International Finance I (Graduate), Kyu Ho Kang*

HONORS AND GRANTS

2026	Graduate School Professional Development Awards	<i>University of Notre Dame</i>
2026	GSG Conference Presentation Grant	<i>Notre Dame Graduate Student Government</i>
2023–24	BK21 Fellowship	<i>Korea University</i>
2023–24	BK21 Graduate School Mentor–Mentee Program	<i>Korea University</i>
2022	Outstanding Research Proposal	<i>BK21 Educational Research Group of Economics</i>
2022	36th Economic Dissertation Contest 1st Place	<i>Maeil Business Newspaper</i>
2021	Research Support Program (Jin-Ri Scholarship)	<i>Korea University</i>

PRESENTATION

2026	Midwest Econometrics Group (MEG) Annual Meeting, Cincinnati (Scheduled)
	European Seminar on Bayesian Econometrics (ESOB) 2026 Rome
2025	The 3rd Tokyo–Taipei–Seoul Macroeconomics Network Workshop, Seoul
2024	Economics Department PhD Conference, Seoul
2020	Korea–Japan Joint Economics Seminar, Seoul (Webinar)

SKILLS AND LANGUAGES

Program Skills

Matlab, Python, R, STATA, Dynare, L^AT_EX

Languages

Korean (native), English (fluent)

NON-ACADEMIC WORK EXPERIENCE

2017–19 Republic of Korea Army (Mandatory), Finished service as Sergeant

Last updated: October 2, 2026.